The Empirical Model in *photic*: Lyzenga's Log-Linear Fit, Diffuse Attenuation and Depth Errors

Technical summary

20 October 2026

Abstract

`MODEL lyzenga_log_linear` fits Lyzenga's log-linear depth model to soundings, such as ICESat-2 bathymetric photons. The model can be fitted to one image or to several scenes at once. Its introduction derives Lyzenga's model from the radiative transfer equation, through Lee et al.'s semi-analytical approximation, which *photic*'s physics-based model inverts. The note then sets out how *photic* makes three estimates.

1. The depth, and its 1σ prediction error (`ERROR`), from a robust weighted least-squares fit.
2. The diffuse attenuation coefficient K of every scene and band, as a smooth field over the scene (`ATTENUATION`, stage 1). It is estimated at the soundings, using the physical model that Lyzenga's regression linearises.
3. A physical depth at every cell, from how the attenuation differs between bands and scenes, with its own error estimate (stage 2).

All of it is implemented in C, in `model/lyzenga.c`. The results quoted are from the Arabian Gulf validation site between Dubai and Abu Dhabi: six Landsat 8 scenes, against ICESat-2 ATL24 split by pass into calibration and test halves.

Lyzenga's fit across all scenes jointly scores 90.0% hits on the test half. Its error estimate holds 80% of the errors within $\pm1\sigma$, but its tails are heavier than Gaussian. K at 443–561 nm is plausible and spatially coherent. K at 655 nm falls below pure water's absorption, so it is not physical. The depth from the attenuation is a diagnostic, not a product, and its error estimate is too small by a factor of about two to three.

Lyzenga et al.'s calibration condition (their eq. 12) explains where the fit is unbiased. The stage-1 attenuation shows that the free fit meets it only at 5–10 m, and that the saturation near 16 m is the condition failing as $\ln L$ flattens. Imposing it on the fit changes almost nothing. Dividing each predictor by the local attenuation (`NORMALISE`) gains about 3 points of hits on the same cells, nearly all at 10–20 m, but loses coverage, so the chosen model is unchanged.

Lyzenga's 1978 paper adds two things. Dividing the error by the local gain (its eq. 17) makes σ grow where the fit saturates: beyond 15 m, 87% of the test errors now lie within $\pm1\sigma$, against 23%. Its depth-invariant bottom indices, computed with the local attenuation, remove most of the depth from the signal but not all, and do not yet give a consistent bottom map at the Gulf. Asking for one bottom in every scene (`CONSISTENT`) recovers the depth exactly on synthetic data, but not at the Gulf, where depth explains only about half of the scenes' differences.

Contents

1 Introduction: from radiative transfer to Lyzenga's model **2**
1.1 The physics: radiative transfer over a reflecting bottom 3
1.2 Lee's semi-analytical approximation 3
1.3 Lyzenga's model as an approximation to Lee's 4

2 Notation and inputs **5**

3 The reference and the training soundings **6**

4 Lyzenga's regression **6**
4.1 The model and its fit . . . 8
4.2 Rescaling against regression dilution (`POWER`) . . . 8
4.3 The depth error estimate (`ERROR`) . . . 8

5 The attenuation behind the regression **10**

6 Stage 1: the attenuation coefficients at the soundings **13**
6.1 A smooth field for each grid . . . 13
6.2 Weights from the noise . . . 13
6.3 Selecting soundings by the fitted signal . . . 14
6.4 Outputs . . . 14

7 Stage 2: a depth from the attenuation **14**
7.1 A straight line at every cell . . . 14
7.2 The depth error estimate . . . 15

8 Results at the Gulf **16**
8.1 The attenuation coefficients . . . 16
8.2 The depth from the attenuation . . . 16

9 Lyzenga's calibration condition **17**
9.1 The condition . . . 17
9.2 What the chosen fit does . . . 17
9.3 Imposing it . . . 18
9.4 Results . . . 18

10 Depth-invariant bottom indices **19**

11 One bottom in every scene: a negative result **20**

12 Usage **22**

1 Introduction: from radiative transfer to Lyzenga's model

photic has two kinds of depth model. Its semi-analytical model (`MODEL Lee_Semi_Analytical`, `paper/photic.tex`) inverts Lee et al.'s approximation to the physics for depth, water and bottom at every cell. The empirical model of this note fits Lyzenga's log-linear regression to soundings. This section sets out how the two are related: the physics, Lee's approximation to it, and Lyzenga's model as a further approximation to Lee's. The approximations made along the way are the ones the rest of this note tests at the Gulf.

1.1 The physics: radiative transfer over a reflecting bottom

In a horizontally uniform water column, the radiance $L(z,\xi)$ at depth z (positive down) in direction ξ obeys the radiative transfer equation (Mobley 1994)

$$\mu \frac{\mathrm{d}L(z,\xi)}{\mathrm{d}z} = -c(z)\,L(z,\xi) + \int_{4\pi} \beta(z;\xi' \to \xi)\,L(z,\xi')\,\mathrm{d}\Omega', \qquad c = a + b, \tag{1.1}$$

where μ is the cosine of the direction's zenith angle, a and b are the absorption and scattering coefficients, and β is the volume scattering function (inelastic scattering and internal sources are left out). Two boundary conditions close it:

- at the surface, sunlight and skylight are refracted into the water, and upwelling light is partly reflected back down (Fresnel, at a wind-roughened interface);
- at the bottom, depth Z, a Lambertian bottom of irradiance reflectance (albedo) ρ returns $L = \rho\,E_d(Z)/\pi$ in every upward direction, where E_d is the downwelling irradiance.

After atmospheric correction, the satellite gives the remote-sensing reflectance $R_{rs} = L_w/E_d(0^+)$: the water-leaving radiance over the downwelling irradiance, just above the surface. Codes such as Hydrolight and OSOAA solve (1.1) numerically, and the validation benchmark of *photic* uses OSOAA as its exact reference (tier 2). Inverting it at every cell would need a, b and β at every wavelength, which is far more than a few bands can determine, and each forward solution is expensive.

1.2 Lee's semi-analytical approximation

Lee et al. (1998, 1999) fitted a closed form to Hydrolight solutions of (1.1), following Gordon et al. (1988) for optically deep water and Maritorena et al. (1994) for the bottom's contribution. The subsurface reflectance $r_{rs} = L_u(0^-)/E_d(0^-)$ is related to the measured one by

$$R_{rs} = \frac{0.5\,r_{rs}}{1 - 1.5\,r_{rs}}, \tag{1.2}$$

and is the sum of a water-column and a bottom term:

$$r_{rs} = r_{rs,\infty}\left(1 - e^{-M_C \kappa Z}\right) + \frac{\rho}{\pi}\,e^{-M_B \kappa Z}, \qquad M_{C,B} = \sec\theta_{w,\mathrm{sun}} + D_{C,B}\,\sec\theta_{w,\mathrm{view}}. \tag{1.3}$$

Every quantity but the angles and Z depends on the band:

- $\kappa = a + b_b$ is the attenuation (written k in `paper/photic.tex`), with b_b the backscattering coefficient;
- $u = b_b/(a + b_b)$, and the deep-water reflectance is $r_{rs,\infty} = (0.084 + 0.170\,u)\,u$;
- $D_C = 1.03\sqrt{1 + 2.4\,u}$ and $D_B = 1.04\sqrt{1 + 5.4\,u}$ lengthen the upward path for photons scattered by the water column and reflected by the bottom;
- θ_w are the zenith angles in water.

The water's a and b_b are then written in three parameters (phytoplankton P, gelbstoff G, particle backscatter X) and the bottom as a mixture of library spectra, so each cell has a few unknowns besides Z. *photic*'s semi-analytical model is (1.2) and (1.3) (`samodel_Rrs` in `model/samodel.c`), inverted across several scenes at once, with each scene's water column $Z = H + h_s$ for depth H below the datum and tide h_s.

1.3 Lyzenga's model as an approximation to Lee's

Rearranging (1.3), the signal above deep water is

$$r_{rs} - r_{rs,\infty} = \frac{\rho}{\pi}\, e^{-M_B \kappa Z} - r_{rs,\infty}\, e^{-M_C \kappa Z}. \tag{1.4}$$

Three approximations turn this into Lyzenga's model.

1. **One path length.** With $D_C \approx D_B$, so $M_C \approx M_B = M$, the two exponentials merge:

$$r_{rs} - r_{rs,\infty} \approx \left(\frac{\rho}{\pi} - r_{rs,\infty}\right) e^{-gZ}, \qquad g = M\kappa. \tag{1.5}$$

   This is Lyzenga's (1978) model: a single exponential in depth, from an *apparent* bottom reflectance $\rho/\pi - r_{rs,\infty}$, the bottom less the water column's own reflectance (his eq. 6, Section 5). In clear water the approximation is close ($u \to 0$: $D_C = 1.03$, $D_B = 1.04$); it weakens as backscatter grows ($u = 0.1$: $D_C = 1.15$, $D_B = 1.29$). Over a bright bottom the first term of (1.4) dominates and $g \approx M_B\kappa$.

2. **A linear surface.** By (1.2),

$$R_{rs} - R_{rs,\infty} = \frac{0.5\,(r_{rs} - r_{rs,\infty})}{(1 - 1.5\,r_{rs})(1 - 1.5\,r_{rs,\infty})}.$$

   For r_{rs} up to about 0.05 the denominator moves by less than 8%, so above the surface $\ln(R_{rs} - R_{rs,\infty})$ is the log of (1.5) plus a near-constant. A spectrally flat offset in R_{rs} (residual glint, Δ in `paper/photic.tex`) cancels in the difference if deep water has it too.

3. **A bottom of fixed colour and varying brightness.** If, across the area fitted, $\rho_m/\pi - r_{rs,\infty,m} = A(\mathbf{x})\, r_m$ for band m (bottoms differ in brightness A but share a colour r_m), and the water's κ_m and $r_{rs,\infty,m}$ are constant, then

$$\ln(R_{rs,m} - R_{rs,\infty,m}) \approx B(\mathbf{x}) + c_m - g_m Z, \qquad B = \ln A. \tag{1.6}$$

   This is (5.4). The factor is exact only where the bottom is much brighter than deep water; otherwise a change of bottom also changes the colour of the apparent reflectance.

Each band then gives one linear equation in the two unknowns B and Z. Two bands eliminate the brightness:

$$Z = \frac{(c_1 - c_2) - (X_1 - X_2)}{g_1 - g_2}, \qquad X_m = \ln(R_{rs,m} - R_{rs,\infty,m}), \tag{1.7}$$

which is Lyzenga's regression (4.1) with $a_2 = -a_1 = 1/(g_1 - g_2)$. With n bands, any coefficients with $-\sum_m a_m g_m = 1$ give the depth, and $\sum_m a_m = 0$ makes it independent of the brightness; these are Lyzenga et al.'s (2006) eqs. 12 and 13 (Section 9). The regression does not need g_m or c_m: it learns them, in the combination it needs, from the soundings. The price is that they must hold over the whole area and every bottom it covers.

The derivation says what to expect when the approximations fail.

- **Varying water.** g_m changes across the scene and between scenes, so one set of coefficients is calibrated only on average. Stage 1 of `ATTENUATION` (Section 6) estimates $g_m(\mathbf{x})$ as a smooth field instead.

- **The plain $\ln L$.** Without the deep-water signal subtracted, $\ln r_{rs} = \ln\bigl(r_{rs,\infty} + (\rho/\pi - r_{rs,\infty})e^{-gZ}\bigr)$ by (1.5), whose slope with depth, $-g\,(r_{rs} - r_{rs,\infty})/r_{rs}$, falls to zero towards deep water. This is the saturation of Section 9.

- **The attenuation it recovers.** `ATTENUATION` writes $g_m = K_m M_s$ with $M_s = \sec\theta_{w,\text{sun}} + \sec\theta_{w,\text{view}}$, so under Lee's model $K_m \approx \kappa_m(\sec\theta_{w,\text{sun}} + D_B \sec\theta_{w,\text{view}})/M_s \geq \kappa_m$. Since $\kappa_m \geq a_w + b_{bw}$, a K below pure water's absorption (Section 8) is not physical.

- **Dark bottoms.** Where $\rho/\pi < r_{rs,\infty}$ the apparent reflectance is negative, the signal lies below deep water's, and the logarithm is undefined; those cells are dropped (`DEEP`).

The rest of this note follows the empirical side: the inputs and the reference (Sections 2 and 3), Lyzenga's regression and its error (Section 4), the attenuation behind it in two stages (Sections 5–7), results at the Gulf (Section 8), Lyzenga et al.'s calibration condition (Section 9), the depth-invariant bottom indices (Section 10), one bottom in every scene, a negative result (Section 11), and usage (Section 12).

2 Notation and inputs

Grids. The command takes n band grids L_m, $m = 1, \ldots, n$, all on one grid. They can be remote-sensing reflectance from ACOLITE, or radiance. With `ATTENUATION ... SCENES` S, the grids are scene-major: S scenes of $n_b = n/S$ bands each, in the same band order. Grid m belongs to scene $s(m) = \lfloor m/n_b \rfloor$.

Window means. Each band is averaged over a $(2r-1) \times (2r-1)$ window of valid cells (`RADIUS` r), giving $\bar{L}_m(\mathbf{x})$. A cell is modelled only where every band is valid. `LAND` and `SHALLOW` masks, if given, further restrict it.

Deep-water signal and noise. For each band:

- $L_{\infty,m}$ is either given (`LSM`) or the p-th percentile of the window means (`PERCENTILE`, default $p = 1$).

- σ_m is the robust spread of the darkest cells: 1.4826 times the median absolute deviation of the window means at or below that percentile. It stands for the noise of a deep-water cell.

The signal above deep water is

$$v_m(\mathbf{x}) = \bar{L}_m(\mathbf{x}) - L_{\infty,m}. \tag{2.1}$$

A band sees the bottom while $v_m \geq k\,\sigma_m$ (`DEEP` k, default 2). A cell where any band fails this test is optically deep for the Lyzenga fit, and gets no depth.

Soundings and tides. Soundings z are depths below the datum, positive down. Each sounding goes to its nearest cell; only soundings within `DEPTHS` $[z_{\min}, z_{\max}]$ are used. `TIDE` h is the water level above the datum when the image was taken, so the light saw a water column $Z = z + h$. With several scenes, scene s has its own level h_s (`TIDES`), and $Z_s = z + h_s$. Outputs are depths below the datum, negative down.

The soundings file must hold depths below the datum. If the tide is added both to the file and through `TIDE`, it is counted twice; this happened in the Gulf composite runs and was fixed on 20 October 2026.

3 The reference and the training soundings

The reference. ICESat-2 ATL24 v002 seabed photons over the box, as depths below MSL, are binned to the image's cells (0.0003°): a cell is the median of at least 3 photons spreading at most 0.5 m (Figure 1a). The passes are split at random into two halves, so the halves are separate tracks: 44 passes calibrate (6,238 cells) and 43 test (5,695 cells). A test cell that also holds a calibration photon is dropped. Settings are chosen by 4-fold cross-validation by pass within the calibration half, and the test half is scored once.

The training filter. Some cells plainly disagree with the image. The filter compares each calibration cell with Stumpf's blue/green log ratio, $\ln(1000R_{483})/\ln(1000R_{561})$ of the median composite, over 3×3 cells (Figure 2). It takes the running median of depth against the ratio and its robust spread (1.4826 MAD, at least 0.5 m), in 40 equal-count bins; the Spearman correlation is 0.89. The two directions differ.

- **Shallower than the ratio implies:** an ICESat-2 seabed where the image sees deep water. These come in runs along a few tracks, which suggests misclassified photons along stretches of a pass. They are dropped beyond 3 spreads, everywhere.
- **Deeper than the ratio implies:** nearly all beyond 10 m, where the ratio saturates (its median flattens at 11–13 m). These are probably real, and dropping them would teach the model that deep water does not exist. They are dropped beyond 3 spreads only where the ratio is not saturated (below its 90th percentile).

In the saturated range the spread is held at its value at the saturation point, as the shallow outliers there would otherwise inflate it (to 9.9 m in the top bin) and hide themselves. The filter drops 377 of the 6,238 calibration cells (6.0%): 276 shallower, 101 deeper.

Each fold's filter is fitted on that fold's own training cells. The held-out cells and the test half are never filtered, as the filter uses the image the model sees, and a filtered test would remove exactly the cells the model gets wrong (Table 1).

Table 1: The training filter, for the model chosen before it (joint, four bands, radius 2, `POWER`). The last column is optimistic: it scores only the test cells the filter keeps.

	Out of fold	Test half	Test RMSE (m)	Test, filtered cells
No filter	83.9%	90.0%	1.21	90.0%
Drop shallower than the ratio	84.2%	90.2%	1.19	91.7%
Both ways, not where saturated	**84.6%**	**90.2%**	1.19	92.2%

The filter barely changes the honest scores (+0.7 points out of fold, +0.2 on the test half), and no depth bin is more than a point worse: Huber weighting already discounts these cells in the fit. It is adopted for the training soundings from 21 October 2026 (Figure 1b). With it, cross-validation's top four settings are within 0.2 points (84.6–84.8%), and its choice moves within that tie to radius 3 without `POWER`, which scores 90.0% on the test half (RMSE 1.20 m). Figure 3 is that model against ICESat-2, and Figure 4 is its bathymetry, with its error estimate (Section 4.3). The analyses of Sections 4.3, 8 and 9 used the earlier choice (radius 2, `POWER`) on the unfiltered training set.

4 Lyzenga's regression

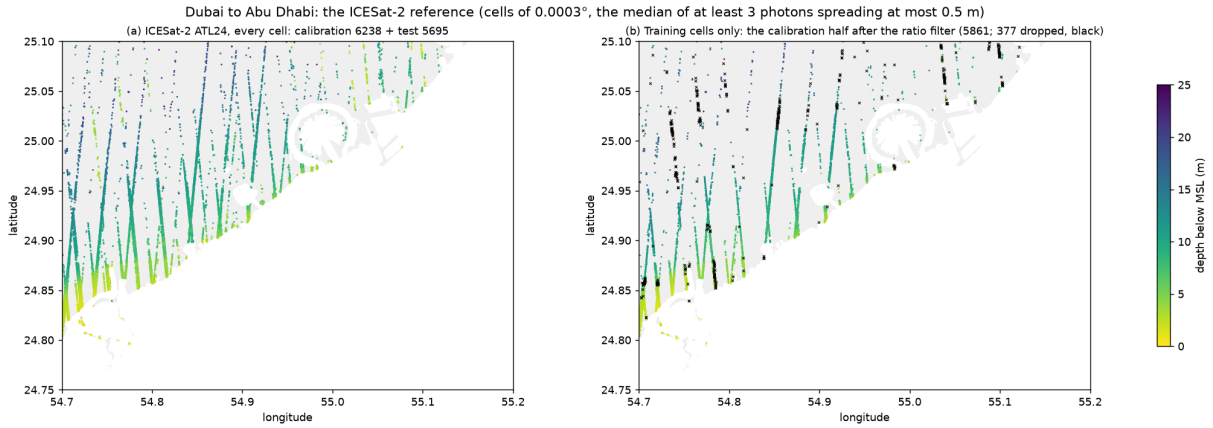


Figure 1: The ICESat-2 reference by depth: (a) every cell, both halves; (b) only the training cells, the calibration half after the ratio filter, with the dropped cells as black crosses.

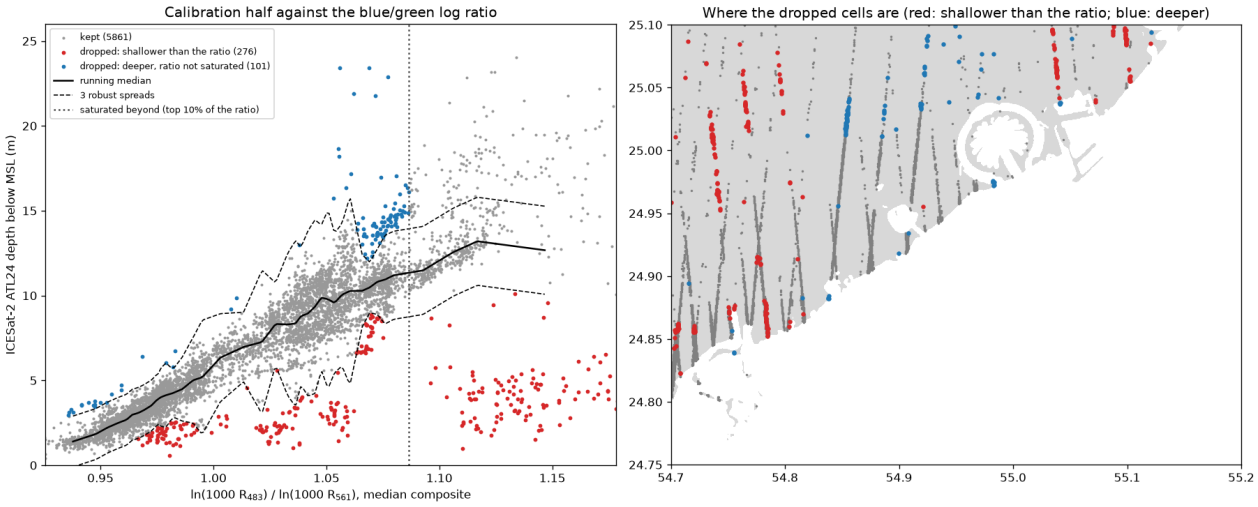


Figure 2: The calibration half against the blue/green log ratio: kept, dropped as shallower than the ratio implies (red) and dropped as deeper where the ratio is not saturated (blue), with the running median and 3 robust spreads; and where the dropped cells are.

4.1 The model and its fit

With $X_m = \ln v_m$, the model (Lyzenga 1978, 1985; Lyzenga et al. 2006) is

$$\hat{Z} = a_0 + \sum_{m=1}^{n} a_m X_m = \mathbf{a}^{\mathsf{T}}\mathbf{x}, \qquad \mathbf{x} = (1, X_1, \ldots, X_n)^{\mathsf{T}}. \tag{4.1}$$

Setting `LSM 0` gives the plain $\ln L$, with no deep-water signal subtracted; Lyzenga et al. (2006) always subtract one. Fitted to every scene's bands at once, equation (4.1) is the *joint* model. It includes every weighted sum of per-scene fits as a special case. The scenes' tides are constants per scene, so they go into the coefficients, and the target is depth below the datum (`TIDE 0`).

The rows are the N soundings whose cells see the bottom in every band. The coefficients minimise

$$\sum_{i=1}^{N} w_i \big(Z_i - \mathbf{a}^{\mathsf{T}}\mathbf{x}_i\big)^2, \tag{4.2}$$

solved by Householder QR of $W^{1/2}A$, which gives the triangular factor R with $R^{\mathsf{T}}R = A^{\mathsf{T}}WA$. The weights are $w_i = w_i^0\,\psi_i$:

- w_i^0 is 1, or 1/(count in the sounding's depth bin) with `BALANCE`.
- ψ_i are Huber weights, from iteratively reweighted least squares: $\psi_i = \min\big(1,\ k_H s/|e_i|\big)$, where e_i is the residual, $s = 1.4826\,\mathrm{median}|e_i|$, and $k_H = 1.345$ (`HUBER`). The iterations start from the least-squares solution.

4.2 Rescaling against regression dilution (`POWER`)

Noise in the predictors shrinks the fitted depths towards the mean, so deep water reads too shallow. `POWER` uses classical calibration to correct this:

1. At the soundings, the fit's depths are modelled as $\hat{Z} = c\,Z^d$, fitted robustly in logs: $\ln\hat{Z} = \ln c + d\ln Z$, with the same Huber IRLS.
2. The law is inverted: $Z' = (\hat{Z}/c)^{1/d}$.

By default, the inverse is applied only beyond the crossover $\hat{Z}_\times = c^{1/(1-d)}$. That is where the fit is compressed into too little depth, so shallow depths keep the fit's smaller scatter. `POWER ALL` applies the inverse at every depth. At the Gulf, the joint fit gives $\hat{Z} = 1.081\,Z^{0.959}$, with the crossover at 6.9 m.

4.3 The depth error estimate (`ERROR`)

The `ERROR` grid is the 1σ prediction error of a new depth at cell $\mathbf{x}$. Write W for the final weights, rescaled to average 1 ($\tilde{w}_i = N w_i / \sum_j w_j$). The residual variance of a sounding of average weight is

$$s^2 = \frac{1}{N-p}\sum_{i=1}^{N} \tilde{w}_i\, e_i^2, \qquad p = n+1. \tag{4.3}$$

The coefficients' covariance follows from the QR factor:

$$C = s^2\big(A^{\mathsf{T}}\tilde{W}A\big)^{-1} = s^2\,\frac{\sum_j w_j}{N}\,(R^{\mathsf{T}}R)^{-1}. \tag{4.4}$$

The prediction variance at a cell with predictors $\mathbf{x}$ is then

$$\sigma^2_{\hat{Z}}(\mathbf{x}) = s^2 + \mathbf{x}^{\mathsf{T}} C\, \mathbf{x}. \tag{4.5}$$

It has two parts: the scatter of one sounding about the model, and the uncertainty of the coefficients, which grows as $\mathbf{x}$ moves away from the soundings' centre.

Through `POWER`, a small error in $\hat{Z}$ scales by $\mathrm{d}Z'/\mathrm{d}\hat{Z} = Z'/(d\,\hat{Z})$. So where the rescaling applies,

$$\sigma_{Z'} = \sigma_{\hat{Z}}\, \frac{Z'}{d\,\hat{Z}}. \tag{4.6}$$

What it represents. Because Huber weighting is used, s^2 in (4.3) is a robust variance. Outlying soundings are down-weighted, so s (0.70 m at the Gulf) describes the core of the residuals, not their tails (in-sample rms 1.86 m). The estimate assumes

- residuals independent between soundings, with one variance at every depth and place;
- the model's form is right.

It therefore cannot see:

- saturation where the bands stop seeing the bottom;
- spatially correlated error, for example from a change of bottom type;
- errors in the soundings themselves.

Noise and gain (`NOISE`, `GAIN`). Lyzenga (1978) derives the depth error from the sensor noise and the signal's sensitivity to depth C (his eqs. 16 and 17):

$$(\Delta Y_N)^2 = \sum_i A^2_{Ni} \frac{\mathrm{NE}\Delta L_i^2}{(L_i - L_{si})^2}, \qquad \Delta Z = \frac{\Delta Y_N}{C}. \tag{4.7}$$

Two options of `ERROR` follow from them.

- `NOISE` adds each cell's own noise, $n^2(\mathbf{x}) = \sum_m a_m^2 \sigma_m^2/(\bar{L}_m - L_{0,m})^2$, where $L_{0,m}$ is the signal subtracted in the fit (0 for $\ln L$). The soundings' mean of it, $\bar{n}^2$, is already part of s^2, so it is taken out first: $\sigma^2 = \max(s^2 - \bar{n}^2, 0) + n^2(\mathbf{x}) + \mathbf{x}^{\mathsf{T}} C \mathbf{x}$.
- `GAIN` (with `ATTENUATION`) divides σ by the local gain of the fitted depth on the true depth. This is C, and it is the gain of Section 9:

$$G(\mathbf{x}) = -\sum_m a_m\, g_m(\mathbf{x})\, f_m, \qquad f_m = \min\Big(1, \frac{\bar{L}_m - L_{\infty,m}}{\bar{L}_m - L_{0,m}}\Big), \tag{4.8}$$

  where $L_{\infty,m}$ is the deep-water signal (the percentile), so $f_m = 1$ for $\ln(L - L_\infty)$ and falls towards deep water for $\ln L$. G needs no depth: it comes from the observed signal and the K maps. Where the fit saturates, G falls and σ grows. G is capped at 1, so the shallow water keeps the fit's own scatter, and floored at 0.05. With `POWER`, the larger of its rescaling and $1/G$ is used, as both undo the same compression.

Calibration at the Gulf. This is the joint model, four bands per scene, with `POWER`, scored against ICESat-2 cells that the fit did not use (Table 2).

- **Plain `ERROR`:** 80% of the errors lie within $\pm 1\sigma$ and 93.6% within $\pm 2\sigma$ (68 and 95% for a Gaussian), and the rms of e/σ is 1.6. So the core is conservative and the tails are heavy. σ is 0.70–0.76 m at every depth, because $\mathbf{x}^{\mathsf{T}} C \mathbf{x}$ is small next to s^2 for 6,017 soundings. It misses the saturation entirely: beyond 15 m, only 23% of the errors are within $\pm 1\sigma$.
- **With `NOISE GAIN`:** 86.9% lie within $\pm 1\sigma$ and 96.4% within $\pm 2\sigma$, and the rms of e/σ is 1.14. At 15–20 m, 87% are within $\pm 1\sigma$, with a median σ of 3.1 m. Nothing changes shallower than 5 m. The noise itself is small, 0.13–0.18 m of the fit's 0.65–0.75 m; the gain does the work.
- **Out of fold** (the four cross-validation folds, calibration half), the same holds: the share within $\pm 1\sigma$ beyond 15 m goes from 9 to 81%, and the rms of e/σ from 2.9 to 1.6. The two forms of the gain (capped at 1 or not) were compared on the test half first, so this is the confirmation.
- **Still too small at 20–30 m** (22%): there the depth no longer responds at all, and the error is most of the depth.

So with `NOISE GAIN`, σ is a fair guide to the error at every depth the model reaches, and it flags where it stops reaching. It is still not a bound.

Table 2: Calibration of Lyzenga's `ERROR` on the Gulf's test half (joint model, four bands, `POWER`; 5,543 cells with a depth), plain and with `NOISE GAIN`.

		`ERROR`		`ERROR NOISE GAIN`		
Depth (m)	Cells	Within $\pm 1\sigma$	Median σ	Within $\pm 1\sigma$	Median σ	RMSE (m)
0–2	161	83%	0.70	83%	0.69	2.38
2–5	1,352	87%	0.70	87%	0.69	1.32
5–10	2,027	79%	0.74	82%	0.75	0.78
10–15	1,737	84%	0.75	94%	0.96	0.65
15–20	243	23%	0.76	87%	3.08	2.15
20–30	23	0%	0.76	22%	3.39	8.77
All	5,543	80%	0.74	87%	0.75	1.21

5 The attenuation behind the regression

Lyzenga's model is the linearisation of a two-flow model of the water-leaving signal over an optically shallow bottom, which Section 1.3 derives from Lee's:

$$L_m - L_{\infty,m} \;\approx\; A(\mathbf{x})\, r_m \, \exp\!\big(-g_m(\mathbf{x})\, Z_s\big), \qquad g_m = K_m\, M_s, \tag{5.1}$$

where

- $A(\mathbf{x})$ is the bottom's brightness, taken as spectrally flat in logs (Lyzenga's assumption);
- r_m is the band's bottom colour, together with the band's deep-water and surface constants;

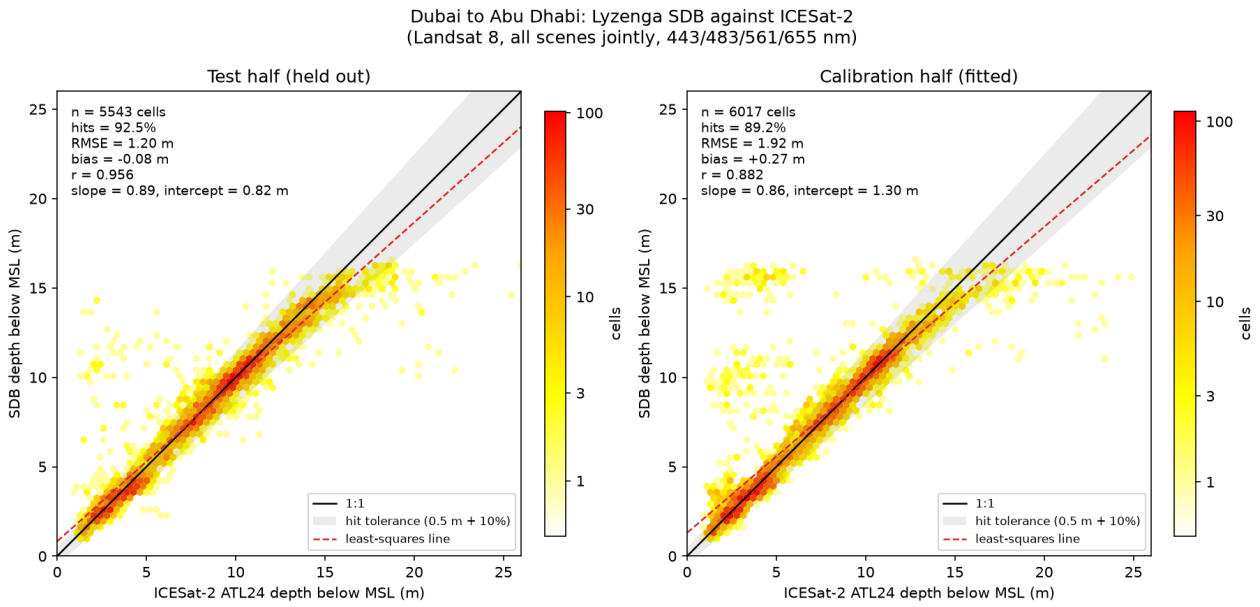


Figure 3: The joint Lyzenga model (four bands, radius 3, trained on the filtered cells; Section 3) against ICESat-2 at the Gulf, as density of cells (white: none). The test half is held out by pass. The calibration panel shows every calibration cell, though the fit used only those the filter keeps. The third panel pools both halves (11,560 cells: 90.8% hits, RMSE 1.61 m); since half of it was fitted, it is not an out-of-sample score. Hits here count cells with a depth. Counting every reference cell, the test-half score is 90.0%.

- K_m is the effective diffuse attenuation of the band;
- M_s is the path length in water per metre of depth, down and back up.

M_s comes from the sun and view zeniths in air (`SUNZENITH`, `VIEWZENITH`), refracted into water with $n_w = 1.34$:

$$M_s = \sec\theta_{w,\mathrm{sun}} + \sec\theta_{w,\mathrm{view}}, \qquad \sin\theta_w = \sin\theta / n_w. \tag{5.2}$$

With both zeniths at 0, $M_s = 2$. For the Gulf scenes (sun zenith 35–51°, near-nadir view), M_s is about 2.1–2.25.

If the down- and up-welling attenuations differ, $g_m = K_d \sec\theta_{w,\mathrm{sun}} + K_u \sec\theta_{w,\mathrm{view}}$. The reported $K_m = g_m / M_s$ is then their path-weighted mean.

Lyzenga (1978, eq. 6) shows that scattering in the water column has the same depth dependence as the bottom's signal. So $A\,r_m$ is an *apparent* bottom reflectance,

$$r'_B = r_B - \frac{\pi\beta(\mu_s)}{K(\mu + \mu')}, \tag{5.3}$$

which depends on the water as well as the bottom (β is the volume scattering function, and μ, μ' the cosines of the paths). A bottom darker than the water column's own scattering has $r'_B < 0$: its signal lies below deep water's, $L < L_\infty$, and its cell is dropped. The same paper notes the model's remaining nonlinearity: internal reflection at the surface bends $\ln(L - L_\infty)$ against depth in very shallow water over bright bottoms.

Taking logs of (5.1) gives, for grid m of scene s,

$$X_m = \ln v_m = B(\mathbf{x}) + c_m - g_m(\mathbf{x})\, Z_s, \qquad B = \ln A,\ c_m = \ln r_m. \tag{5.4}$$

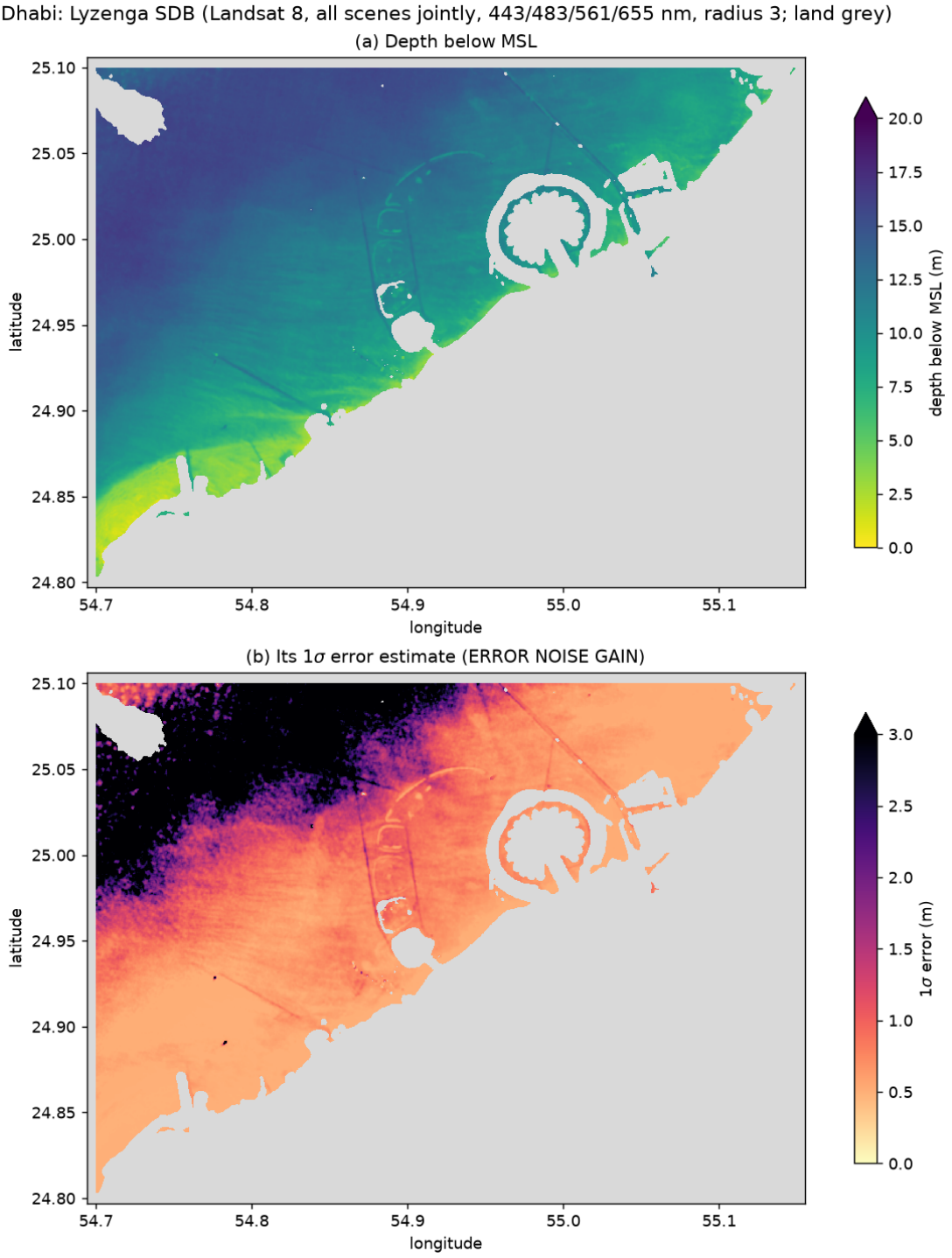


Figure 4: The satellite-derived bathymetry at the Gulf: the joint Lyzenga model of Figure 3 over all 724,111 water cells. (a) The depth below mean sea level. (b) Its 1σ error estimate, `ERROR NOISE GAIN` (Section 4.3): median 0.75 m, 90th percentile 4.3 m. The depth reaches at most 16.5 m, though ICESat-2 finds water to 28 m. Towards the north-west the signal no longer responds to depth (the saturation of Section 9), and the error estimate rises past 3 m to flag it.

With g_m and c_m constant and B constant, Lyzenga's regression (4.1) is this equation solved for Z. In practice g_m varies over a scene, and between scenes, with the water. `ATTENUATION` estimates it explicitly, in two stages.

In stage 1, $L_{\infty,m}$ is the `LSM` value where that is positive, and otherwise the percentile. So `LSM 0` (the plain $\ln L$) still subtracts a deep-water signal for the attenuation.

6 Stage 1: the attenuation coefficients at the soundings

6.1 A smooth field for each grid

At the soundings the depth is known, so for each grid m separately, equation (5.4) is fitted with B left in the residuals:

$$X_{m,i} = c_m - g_m(\mathbf{x}_i)\, Z_{s,i} + \varepsilon_{m,i}. \tag{6.1}$$

$g_m(\mathbf{x})$ is a bilinear interpolant of values $\theta_{m,q}$ on a regular grid of nodes, set `NODES` degrees apart (default 0.05°; 12×9 nodes over the Gulf box):

$$g_m(\mathbf{x}) = \sum_{q=1}^{4} f_q(\mathbf{x})\, \theta_{m,q(\mathbf{x})}, \tag{6.2}$$

where $q(\mathbf{x})$ are the four nodes around the cell and f_q their bilinear weights. The unknowns, c_m and the Q node values $\theta_{m,\cdot}$, minimise

$$\sum_i w_{m,i}\Big(X_{m,i} - c_m + Z_{s,i}\sum_q f_q(\mathbf{x}_i)\theta_{m,q}\Big)^2 \;+\; \lambda \sum_{(q,q')} \big(\theta_{m,q} - \theta_{m,q'}\big)^2. \tag{6.3}$$

The second sum runs over neighbouring pairs of nodes, east–west and north–south. It is a first-difference smoothing penalty, which makes the field smooth and fills nodes without soundings. Its weight is scaled to an average node's data:

$$\lambda = w_s\, \frac{\sum_i w^0_{m,i} Z^2_{s,i}}{Q}, \tag{6.4}$$

where w_s is `SMOOTHING` (default 1; 0 requires soundings near every node). The normal equations, of size $Q+1$, are symmetric positive definite and are solved by Cholesky factorisation. The residuals are robustified by the same Huber IRLS as in Section 4, with scale $1.4826\,\mathrm{median}|\varepsilon|$.

6.2 Weights from the noise

The noise σ_m on v_m gives $\ln v_m$ a variance of about σ_m^2/v_m^2 (delta method). This grows without limit as the signal sinks towards the deep-water level. A floor $\epsilon = 0.05$ stands for the model's own error in $\ln v$. The weights are

$$w^0_{m,i} = \Big(\frac{\sigma_m^2}{v_{m,i}^2} + \epsilon^2\Big)^{-1}, \tag{6.5}$$

so a band counts fully while it sees the bottom clearly, and fades out as it loses it. The Huber factors multiply w^0.

6.3 Selecting soundings by the fitted signal

A sounding enters grid m's fit only while its signal is above the noise, $v \geq k\sigma_m$. Selecting, and weighting, by the *observed* signal truncates the data on the dependent variable. At depth, only the cells that happen to be bright pass the test, which flattens the slope and makes K too small. The effect is largest in the red, which loses the bottom within a few metres.

So after the first pass, each candidate sounding is reselected and reweighted by the *fitted* signal instead:

$$\hat{v}_{m,i} = \exp\big(c_m - g_m(\mathbf{x}_i)\, Z_{s,i}\big), \qquad \text{active if } \hat{v}_{m,i} \geq k\sigma_m, \tag{6.6}$$

with $\hat{v}$ in place of v in (6.5). The fit is repeated until the selection stops changing, for at most 10 passes. This raised $K(655)$ at the Gulf by 20–40%. A grid needs at least 10 active soundings.

6.4 Outputs

- $K_m(\mathbf{x}) = g_m(\mathbf{x})/M_{s(m)}$ at every valid cell, as grid `/prefix/_/grid/`.
- `attenuation.csv`: K of every grid at every node, with the node's support (the sum of the bilinear weights of the soundings near it).
- The printout gives, for each grid: c_m, the median K over nodes with support of at least 1, its range, the robust scatter of (6.1), and the number of passes.

Caveat. B is left in the residuals. K is therefore unbiased only where the bottom's brightness does not change systematically with depth across the soundings. If seagrass is deeper than sand, the slope also contains the brightness change.

7 Stage 2: a depth from the attenuation

7.1 A straight line at every cell

With c_m and $g_m(\mathbf{x})$ fixed from stage 1, equation (5.4) at a cell, over all its grids, is a straight line in g_m:

$$y_m \equiv X_m - c_m + g_m(\mathbf{x})\, h_{s(m)} = B(\mathbf{x}) - g_m(\mathbf{x})\, H(\mathbf{x}), \tag{7.1}$$

where H is the depth below the datum. The intercept B is the bottom's brightness and the slope is $-H$. The depth comes from how g differs between bands and scenes, which is what makes it independent of B.

The weighted least-squares line through the grids that see the bottom gives

$$\hat{H} = -\frac{S_{gy}}{S_{gg}}, \qquad \hat{B} = \bar{y} + \hat{H}\bar{g}, \tag{7.2}$$

with

$$\bar{g} = \frac{\sum w_m g_m}{\sum w_m}, \quad \bar{y} = \frac{\sum w_m y_m}{\sum w_m}, \quad S_{gg} = \sum_m w_m (g_m - \bar{g})^2, \quad S_{gy} = \sum_m w_m (g_m - \bar{g})(y_m - \bar{y}). \tag{7.3}$$

The weights are (6.5) with the cell's signal. As in stage 1, the first pass selects and weights by the observed signal; later passes use the fitted signal,

$$\hat{v}_m = \exp\big(\hat{B} + c_m - g_m(\hat{H} + h_{s(m)})\big) = v_m \exp\big(\hat{B} - g_m\hat{H} - y_m\big), \tag{7.4}$$

until the selection settles (at most 10 passes). A cell needs at least 3 grids above $k\sigma_m$.

The outputs are three grids:

- `/prefix/_depth`, the depth $-\hat{H}$ (negative down);
- `/prefix/_depth_sigma`, its error σ_H;
- `/prefix/_bottom`, the bottom brightness $\hat{B}$.

Which differences carry the depth. The leverage comes from how g differs among a cell's grids: between bands, and between scenes, where both the water and the light path change. A prototype (out of fold, 443/483/561 nm) splits them.

- **One intercept per band** removes the bottom colour, as a band's colour offset is the same in every scene, and leaves only the differences between scenes. It scores 49.5% hits against 61.0% for the common intercept. With four bands it ties (56.7 against 56.1%), fixing 0–2 m (5 → 39%) but losing at 10–20 m.
- **One K per band shared by the scenes** leaves only the sun angle and the tide. The Gulf's sun zeniths of 35–51° change the path by 6%, and the result has no depth information (4% hits).

So measuring each scene's water makes the differences between scenes informative, unlike the semi-analytical model's free water per scene. But the scenes' own unmodelled differences (atmosphere, adjacency; Section 10) limit them as much as the bottom colour limits the bands.

7.2 The depth error estimate

σ_H is the classical standard error of the slope of a weighted straight line, with the residual scale estimated from the cell's own scatter about the line:

$$\sigma_H = \sqrt{\frac{\mathrm{RSS}}{(n_u - 2)\, S_{gg}}}, \qquad \mathrm{RSS} = \sum_m w_m \big(y_m - \hat{B} + g_m \hat{H}\big)^2, \tag{7.5}$$

where n_u is the number of grids used.

The leverage is S_{gg}, the weighted spread of the two-way attenuation among the cell's grids. It is small. Between 443 and 561 nm, K differs by only 0.01–0.03 m^{-1}, so g spans about 0.02–0.07 m^{-1}, and the scenes' different sun angles add little. A bottom colour that departs from spectral flatness by δ in ln therefore moves the depth by about $\delta/\Delta g$. For $\delta = 0.1$ that is 1.5–5 m, comparable to the depths being estimated. This is Lyzenga's (1978, eq. 18) result for the two-band ratio method, whose noise error is $\left[\sum_i (\mathrm{NE}\Delta L_i/(L_i - L_{si}))^2\right]^{1/2}/((\kappa_1 - \kappa_2) f)$ and grows without limit as the bands' attenuations approach each other.

Equation (7.5) treats the n_u residuals as independent, with a common scale. It omits three things.

1. **Stage 1's uncertainty.** c_m and $g_m(\mathbf{x})$ are taken as exact. Their errors are shared by every cell near the same nodes, so they are systematic, not random, and they do not average down.
2. **The bottom's colour.** A non-flat bottom spectrum gives each band its own offset in y_m. The same offset recurs in every scene, so the residuals are correlated across scenes, and with only a few distinct bands it can be fitted into the slope rather than showing up as scatter.
3. **Few effective degrees of freedom.** The 18–24 grids are only 3–4 distinct bands seen 6 times under similar water. $n_u - 2$ overstates the information.

All three make σ_H too small. Out of fold at the Gulf, only 25% (four bands) and 37% (443, 483 and 561 nm) of the errors lie within $\pm\sigma_H$, against 68% for a calibrated estimate.

Two corrections are possible:

- propagate stage 1's covariance (the Cholesky factor is at hand) into (7.5);
- inflate σ_H by a factor calibrated out of fold.

Neither is built.

8 Results at the Gulf

8.1 The attenuation coefficients

Over the box, K is 0.05–0.09 m^{-1} at 443 nm, 0.07–0.10 at 483 and 0.065–0.105 at 561 (medians over the scenes), and it varies 1.5–2× across the box (Figure 6). The three bands are fitted independently, yet they map the same pattern: low in the south-west, rising towards Jebel Ali and Dubai. The scenes differ: 2019-03-13 is the clearest, 2018-10-04 the least clear. One set of Lyzenga coefficients cannot represent this variation. These maps are the per-site attenuation targets that the learned model's physics decoder needs (`validation/FUTURE_EXPERIMENTS.md`, H4(c)).

Pure water's absorption a_w (Pope and Fry 1997) is a lower bound on K (Figure 5).

- **443 and 483 nm:** K is 4–15 times a_w, as expected where dissolved and particulate matter dominate.
- **561 nm:** K is barely above a_w (0.065–0.105 against 0.062 m^{-1}). 6% of the node values fall below it, most of them in the clearest scene.
- **655 nm:** every value is below a_w (0.08–0.19 against about 0.37 m^{-1}), so none is physical.

The red loses the bottom within a few metres. Its deep-water signal varies (turbidity, adjacency, the atmospheric residual) by far more than the darkest cells' spread σ_m, which sets the threshold. So nearly every sounding stays active, and the slope is flattened. The flattening already reaches the green on clear scenes.

Two remedies are possible, and neither is built:

- fit the deep-water level per grid, or as a field;
- bound K below by a_w.

A pure-water bound would bind at 561 nm as well as at 655.

8.2 The depth from the attenuation

Table 3 scores the depth from the attenuation against Lyzenga's depth, out of fold. The folds are the four ICESat-2 pass groups that chose Lyzenga's settings, within the calibration half only. The harness reproduces Lyzenga's cross-validation score (83.9%).

The depth from the attenuation is not competitive with Lyzenga's. With so little spread in g, a change of bottom type (seagrass against sand) shifts the band ratios enough to move the depth by metres. Lyzenga's regression absorbs the site's bottom–depth relationships into its coefficients; equation (7.1) cannot. Dropping the red removes the unphysical K and helps the shallowest water

Table 3: Out of fold on the Gulf's calibration half (5,833 cells). A hit is an error within 0.5 m + 10% of the depth.

	Hits	RMSE (m)	Bias (m)	0–2	2–5	5–10	10–20
Lyzenga, joint, four bands	83.9%	2.09	+0.26	58%	79%	90%	90%
Attenuation depth, four bands	57.3%	3.14	−0.58	5%	56%	79%	47%
Attenuation depth, 443/483/561	61.4%	2.84	+0.12	34%	54%	69%	69%

most (0–2 m: 5 → 34%). In the shallowest water the straight line of eq. (7.1) is also least exact, because of internal reflection (Section 5). Its error estimate is too small (Section 7.2).

So `ATTENUATION` is kept, off unless asked, with no effect on Lyzenga's fit. K at 443–561 nm is the usable product. The attenuation depth is a diagnostic.

9 Lyzenga's calibration condition

9.1 The condition

Substituting the physical model (5.4) into the regression (4.1) gives (Lyzenga et al. 2006, eq. 11, in our signs)

$$\hat{Z} = \Big(a_0 + \sum_m a_m (B + c_m)\Big) - \Big(\sum_m a_m g_m\Big) Z. \tag{9.1}$$

The estimate is unbiased when the offset vanishes and the *gain* $G = -\sum_m a_m g_m$ is 1 (their eq. 12). With N bands there are $N + 1$ coefficients, so the fit has $N - 1$ degrees of freedom to spare; it can stay calibrated across several bottoms or waters. A change of a spectrally flat bottom brightness B leaves the depth unchanged if $\sum_m a_m = 0$ (their eq. 13, with the same specific attenuation in every band).

Lyzenga et al. do not impose eq. 12. They fit by unconstrained least squares (their eqs. 14 and 15), as *photic* does, and least squares meets it only on average over the soundings.

With $\ln L$. `LSM 0` uses $X_m = \ln L_m$, not $\ln(L_m - L_{\infty,m})$. Its derivative with depth is $-g_m(L_m - L_{\infty,m})/L_m$, which falls to zero as the signal approaches deep water. The local gain is then

$$G_{\text{eff}}(\mathbf{x}, Z) = -\sum_m a_m \, g_m(\mathbf{x}) \, \frac{L_m - L_{\infty,m}}{L_m}, \tag{9.2}$$

and it falls with depth whatever the coefficients.

9.2 What the chosen fit does

For the joint model ($\ln L$, four bands, `POWER`), the stage-1 attenuation gives the gain directly (Table 4).

- **The fit is calibrated only where the soundings are dense (5–10 m).**
- **The saturation near 16 m is eq. 12 failing:** the predicted gain falls to 0.27 at 15–20 m, against an observed slope of 0.18. The K maps predict it with no reference to the test depths.

Table 4: The gain of eq. (9.1) predicted from the stage-1 attenuation, and the observed slope of the fitted depth against ICESat-2 (test half, by depth bin).

	0–2 m	2–5 m	5–10 m	10–15 m	15–20 m
Predicted gain G_{eff}	1.32	1.08	**1.00**	0.77	0.27
Observed slope	0.05	0.65	**1.00**	0.90	0.18

- **Two departures are explained:** `POWER` stretches the depths beyond 6.9 m, which is why 10–15 m is observed above the prediction. Over a 2 m bin, the reference's own scatter dilutes the observed slope, so the shallow bins are not comparable.
- **The gain also varies across the box,** from 1.30 to 1.84 (10th to 90th percentile) with the nominal attenuation, because K varies 1.5–2× and one set of coefficients cannot follow it. At 10 m, a gain off by 15% is a 1.5 m error.
- **The fit is nearly invariant to a flat brightness change:** $\sum_m a_m = -1.27$, so a 10% brighter bottom moves the depth by 0.12 m. It works through large opposing blue and green coefficients (band sums +25.6 and −21.7), as Lyzenga et al.'s (their $h_1 \approx -h_2$), at the cost of noise amplification ($\sum |a_m| = 102$).

9.3 Imposing it

Two options use the stage-1 attenuation, which now runs before the fit.

- `CONSTRAIN GAIN` imposes $-\sum_m a_m \bar{g}_m = 1$, where $\bar{g}_m$ is the mean attenuation over the fit's soundings. `BRIGHTNESS` adds $\sum_m a_m = 0$.
- `NORMALISE` divides each predictor by the local $g_m(\mathbf{x})$. Then $X_m/g_m = (B + c_m)/g_m - Z$, so the gain is $-\sum_m a_m$ at *every* cell, whatever the attenuation there. With `CONSTRAIN GAIN` the condition becomes $\sum_m a_m = -1$, and `BRIGHTNESS` adds $\sum_m a_m/\bar{g}_m = 0$.

The constraints are linear equalities $C\mathbf{a} = \mathbf{d}$, none involving a_0. They are solved exactly: the constraint matrix is reduced to row echelon form, one coefficient per constraint is eliminated, and the reduced problem is fitted with the same Householder QR and Huber weights. With $\mathbf{a} = T\mathbf{a}_r + \mathbf{t}$, the covariance for `ERROR` is $C = TC_rT^{\mathsf{T}}$. `make test` checks that both constraints hold to four decimals.

Both options want $\ln(L - L_\infty)$: with $\ln L$, the effective attenuation changes with depth, so a fixed constraint is mis-specified.

9.4 Results

Out of fold on the Gulf's calibration half (the same four folds that chose the model; Table 5). Subtracting L_∞ turns the optically-deep test on, which removes 4–7% of the cells. Nearly all are deeper than 10 m, and with four bands it is mostly the red's doing. A missing cell counts as a miss, so the table also scores every variant on the cells where all of them have a depth.

- **Imposing eq. 12 changes almost nothing** (−1.4 to +0.4 points). Once L_∞ is subtracted, the free fit already meets it on average (with `NORMALISE`, the free fit's gain at the soundings is 0.92–1.01), and the constraint does as well as `POWER`. Regression dilution is not the main error.

Table 5: Lyzenga's calibration condition imposed on the joint fit, four bands, out of fold (5,833 cells; "same cells": the 5,201 where every variant has a depth).

	Hits	With a depth	Same cells	RMSE (m)	10–20 m
$\ln L$, `POWER` (chosen)	**83.9%**	96.3%	89.2%	1.60	92%
$\ln L$, `CONSTRAIN GAIN`	83.1%	96.3%	88.5%	1.59	90%
$\ln(L - L_\infty)$, free	77.5%	89.2%	86.9%	1.95	79%
$\ln(L - L_\infty)$, `POWER`	77.6%	89.2%	87.0%	1.99	80%
$\ln(L - L_\infty)$, `CONSTRAIN GAIN`	77.9%	89.2%	87.4%	1.90	83%
with `BRIGHTNESS`	76.2%	89.2%	85.4%	1.94	81%
`NORMALISE`	79.9%	89.2%	**89.6%**	1.78	91%
`NORMALISE CONSTRAIN GAIN`	79.5%	89.2%	89.2%	1.79	91%

- `BRIGHTNESS` **hurts every time,** by 1.5–5 points. The bottom is not spectrally flat, and the free fit uses how bottom colour changes with depth at this site.
- `NORMALISE` **is the real effect.** On the same cells it gains 2.7 points over the same $\ln(L - L_\infty)$ fit with four bands, and 3.0 with 443/483/561 nm, nearly all at 10–20 m (79 → 91% and 80 → 93%). So the spatial variation of K is the dominant error at depth. With 443/483/561 nm it is the best fit overall: 82.2% against $\ln L$'s 78.4%.
- **With four bands it only ties the chosen model** on the same cells (89.6 against 89.2%), and loses overall on coverage. Loosening the deep test (`DEEP 1`, `DEEP 0`) recovers coverage but costs more accuracy than it gains.

So the chosen model is unchanged, and `CONSTRAIN` and `NORMALISE` are kept as options, off by default. `NORMALISE` is how per-site K maps enter an empirical depth. Two next steps are possible: normalise $\ln L$ by its effective attenuation $g_m(L_m - L_{\infty,m})/L_m$, which keeps $\ln L$'s coverage; or fill the optically deep cells from the $\ln L$ fit.

Policy. Every future Lyzenga experiment will include Lyzenga et al.'s calibration constraint, eq. 12, as a variant scored alongside the free fit: `CONSTRAIN GAIN`, with `ATTENUATION` and $\ln(L - L_\infty)$. It adds little here, where the free fit already meets eq. 12 on average and most soundings lie where it holds. Elsewhere it may well matter: with fewer or less evenly spread soundings, noisier bands or stronger regression dilution, least squares can miss the condition, and the constraint enforces it from the measured attenuation instead.

10 Depth-invariant bottom indices

Lyzenga (1978) rotates the N log-bands of one scene into $N - 1$ indices that do not depend on depth, and one that does (his eqs. 8 and 9). Within a scene, $X_j = a_j - b_j Z$, with a_j depending on the bottom and $b_j = g_j$. The rotation takes Y_N along $\mathbf{b}$, so $Y_N = B_m - CZ$ carries the depth, and the other axes orthogonal to it, so they carry only the bottom. His Appendix B gives the rotation:

$$Y_i = \sum_{j=1}^{N} A_{ij} X_j, \qquad A_{ij} = \frac{b_{i+1}\, b_j}{\sqrt{S_i S_{i+1}}} \; (j \le i), \quad A_{i,i+1} = -\sqrt{\frac{S_i}{S_{i+1}}}, \quad A_{ij} = 0 \; (j > i+1), \tag{10.1}$$

for $i = 1, \ldots, N-1$, where $S_i = \sum_{k \le i} b_k^2$. With two bands this is the familiar index $(b_2 X_1 - b_1 X_2)/|\mathbf{b}|$, which `COMPUTE BOTTOM` already provides with a fixed attenuation ratio.

`INDICES` (with `ATTENUATION`) computes them for every scene, with $b_j = g_j(\mathbf{x})$ from stage 1. So the rotation follows the water across the scene, rather than assuming one attenuation ratio everywhere. The outputs are `/prefix/_index/i/_s/scene/`. `make test` checks them against the exact values on synthetic data, where the depth term cancels exactly (largest error 2.3×10^{-4}).

At the Gulf (Figure 7). Each scene's index carries that scene's constants ($\sum_j A_{ij} c_j$: the bands' bottom colour, deep-water and atmospheric terms), so the maps show anomalies from each scene's median. Two checks follow.

- **Most of the depth is removed, but not all.** A band's own $\ln(L - L_\infty)$ has $r = -0.80$ to -0.64 with the ICESat-2 depth. With 443/483/561 nm, index 1 has $r = -0.28$ to $+0.14$, and index 2 still $+0.31$ to $+0.51$. Index 2 shows the same near-shore against offshore split in every scene. There are three candidates: errors in K; a bottom that really changes with depth at this site; and the apparent reflectance of Section 5, which depends on the water.

- **The scenes do not agree.** The bottom is the same in every scene, so its indices should be too, but the median correlation between scenes is only 0.17 (index 1) and 0.38 (index 2); 0.19–0.63 with four bands. Index 1 is dominated by large patterns that differ between scenes: the atmosphere, adjacency and the water.

So the rotation is right, but at this site the bottom's contrast is small next to these effects, and the indices are not yet a bottom map. They stay, off by default, as the formalism for two bottom classes in stage 2 ($Y_N = B_m - CZ$ with a class-specific B_m, which is Lyzenga's own route to the depth) and as features for the learned model.

11 One bottom in every scene: a negative result

The indices of Section 10 disagree between scenes, although the bottom is the same in all of them. `ATTENUATION ... CONSISTENT` asks for that directly, to remove whatever depth enters the indices differently in each scene.

The condition. Each scene's indices use that scene's rotation, so the same bottom gives different indices when the water differs. The right condition is one bottom vector $\beta_b(\mathbf{x})$, of any colour, shared by the scenes:

$$X_{sb} = \beta_b(\mathbf{x}) + c_{sb} - g_{sb}(\mathbf{x})\,(H + h_s), \qquad g_{sb} = \bar{g}_b(\mathbf{x}) + \gamma_{sb}(\mathbf{x}), \quad \textstyle\sum_s \gamma_{sb} = 0, \tag{11.1}$$

for band b of scene s. Removing each band's mean over the scenes removes β_b and $\bar{g}_b H$, and leaves only the scenes' differences of attenuation γ_{sb}, times the depth.

- **What it cannot identify.** A bottom that darkens uniformly with depth, $\beta_b + v_b H$, gives the same data as every scene's attenuation shifted by $-v_b$, plus a constant per scene. Only the tides separate the two, and their spread at the Gulf is 0.35 m. So $\bar{g}_b$ stays from stage 1, and depth that enters every scene alike cannot be removed this way. Index 2's near-shore against offshore split, the same in all six scenes, is that mode.

- **The fit.** γ_{sb} is bilinear on stage 1's nodes, and the scenes' constants are fitted with it, over up to 10^5 of the image's cells. H is fixed at the soundings' cells and free at every other. The problem is bilinear, so it is solved by Gauss–Newton with each cell's H eliminated, and a line search. There is no smoothing penalty: one that favours a small γ collapses it along $\gamma \to \lambda\gamma$, $H \to H/\lambda$, which only the soundings resist. Stage 2 then fits one bottom per band at every cell, and `INDICES` adds the one bottom's indices.

Synthetic data. Six scenes with the Gulf's tides and sun angles, three bands, a patchy bottom of three colours, noise 0.01 in ln, and soundings only in the western half:

- the depth's rms error falls from 1.52 m (stage 2's flat bottom) to 0.09 m, and to 0.09 m in the east, where there are no soundings;
- γ's error falls from 49% to 5%.

`make test` (`tests/con_*`) recovers γ to 10^{-6} and the depth to 0.14 mm, against 1.66 m without `CONSISTENT`.

At the Gulf (Table 6) it does not help.

Table 6: Hits out of fold on the calibration half's folds (all held-out cells), the attenuation's depth with a flat bottom and with `CONSISTENT`, and Lyzenga's fit, free and with `CONSTRAIN GAIN`.

	Flat bottom	`CONSISTENT`	Lyzenga	`CONSTRAIN GAIN`
Four bands	61.3%	58.6%	**84.8%**	80.7%
443/483/561 nm	62.5%	56.2%	80.7%	**83.2%**

- **Mixed by depth.** With four bands it fixes 0–2 m (9 to 40%) and 2–5 m (62 to 76%), but loses 10–20 m (64 to 44%). A pull towards stage 1 (`PRIOR`) does not help.
- **The fit does not settle.** After 50 iterations its parameters still move while the objective changes by 10^{-5}: a flat valley that the data do not determine.
- **The model holds only in part.** At the training soundings, where the depth is known, depth explains only 50–64% of each band's differences between scenes. The rest (residual sd about 0.15 in ln: atmosphere, adjacency, the water-dependent apparent reflectance) is as large as the depth signal, and away from the soundings the fit takes it for γH.
- **The indices get worse** (Figure 8). With the consistent attenuation, each scene's indices agree less between scenes (median r 0.20 to 0.11 for index 1, 0.32 to 0.08 for index 2), and scene 4 breaks into blocks the size of the nodes. The one bottom's indices are smooth and correlate less with depth (+0.15 and +0.14, against up to +0.48), but they match no single scene's, so they are not yet a bottom map.

`CONSISTENT [PRIOR w]` stays, off by default and tested, for scenes whose differences are dominated by the water rather than the atmosphere. It would need better atmospheric correction, or a smooth additive residual per scene, which is partly confounded with γH where the depth is smooth.

12 Usage

```
MODEL lyzenga_log_linear GRIDS g1 ... gn BATHYMETRY H [options, any order]
  LSM l1 ... ln | PERCENTILE p    DEEP k    RADIUS r    DEPTHS zmin zmax
  TIDE h    HUBER k    BALANCE bin    POWER [ALL]    ERROR H_error [NOISE] [GAIN]
  ATTENUATION K SCENES S [TIDES h1 ... hS] [SUNZENITH t1 ... tS]
                         [VIEWZENITH t1 ... tS] [NODES deg] [SMOOTHING w]
  CONSTRAIN GAIN [BRIGHTNESS]    NORMALISE    INDICES    CONSISTENT [PRIOR w]
                              (these, and ERROR ... GAIN, need ATTENUATION)
```

`SCENES` must come before the per-scene lists, and the grids are scene-major. The outputs are:

- `H` and `H_error`;
- with `ATTENUATION`, `K_/grid/` for every grid, and `K_depth`, `K_depth_sigma` and `K_bottom`; with `INDICES`, `K_index/i/_s/scene/`, and with `CONSISTENT` also the one bottom's `K_index/i/`;
- the files `model.bam` (the fitted model as applied), `output.csv` (every sounding, its model depth and whether it was used) and `attenuation.csv`.

At the Gulf, `validation/gulf_lyzenga.py` runs it. That script prepares the inputs, chooses the settings by cross-validation, scores the results and draws the figures; the modelling is all in *photic*. `make test` checks the fit and the attenuation on synthetic data that follow the models exactly:

- `tests/lyz_*`: the coefficients and every sounding's depth;
- `tests/att_*`: K at every node, to 3×10^{-6}, the depth at every cell, within 5 mm (the float precision of the band grids), and the depth-invariant indices, within 2×10^{-3}; then both constraints of `NORMALISE CONSTRAIN GAIN BRIGHTNESS`, exactly;
- `tests/con_*`: with `CONSISTENT`, the scenes' differences of attenuation and the depth at every cell, for a bottom whose colour changes from cell to cell.

References

- Gordon, H. R., Brown, O. B., Evans, R. H., Brown, J. W., Smith, R. C., Baker, K. S. and Clark, D. K. (1988). A semianalytic radiance model of ocean color. *Journal of Geophysical Research* 93(D9), 10909–10924.
- Lee, Z., Carder, K. L., Mobley, C. D., Steward, R. G. and Patch, J. S. (1998). Hyperspectral remote sensing for shallow waters. I. A semianalytical model. *Applied Optics* 37(27), 6329–6338.
- Lee, Z., Carder, K. L., Mobley, C. D., Steward, R. G. and Patch, J. S. (1999). Hyperspectral remote sensing for shallow waters. II. Deriving bottom depths and water properties by optimization. *Applied Optics* 38(18), 3831–3843.
- Lyzenga, D. R. (1978). Passive remote sensing techniques for mapping water depth and bottom features. *Applied Optics* 17(3), 379–383.
- Lyzenga, D. R. (1985). Shallow-water bathymetry using combined lidar and passive multispectral scanner data. *International Journal of Remote Sensing* 6(1), 115–125.

- Lyzenga, D. R., Malinas, N. P. and Tanis, F. J. (2006). Multispectral bathymetry using a simple physically based algorithm. *IEEE Transactions on Geoscience and Remote Sensing* 44(8), 2251–2259.
- Maritorena, S., Morel, A. and Gentili, B. (1994). Diffuse reflectance of oceanic shallow waters: influence of water depth and bottom albedo. *Limnology and Oceanography* 39(7), 1689–1703.
- Mobley, C. D. (1994). *Light and Water: Radiative Transfer in Natural Waters*. Academic Press, San Diego.
- Pope, R. M. and Fry, E. S. (1997). Absorption spectrum (380–700 nm) of pure water. II. Integrating cavity measurements. *Applied Optics* 36(33), 8710–8723.

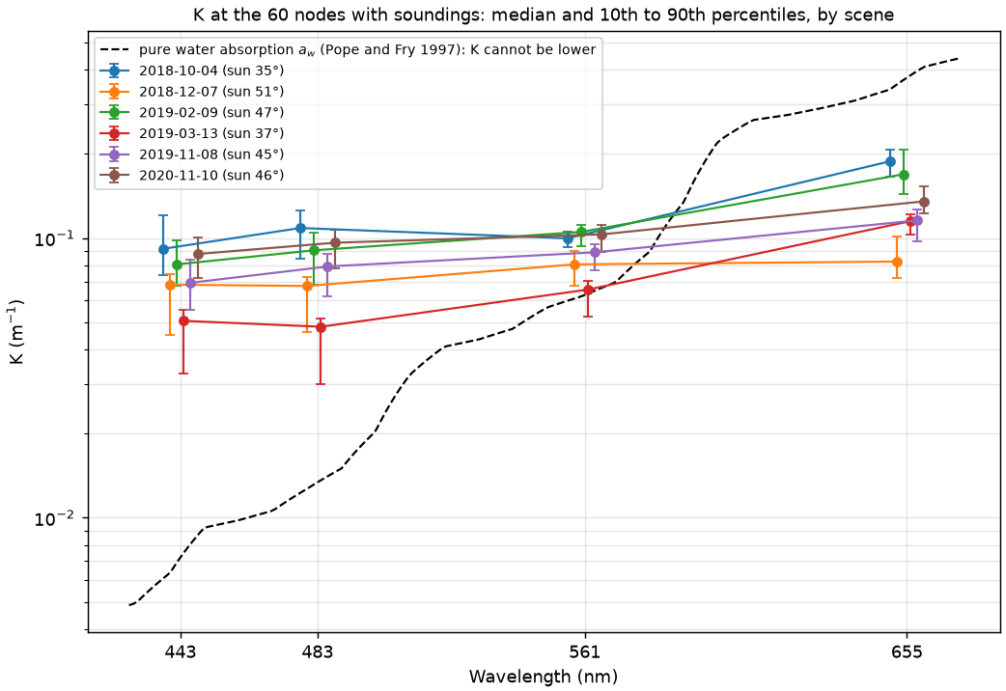


Figure 5: K of each scene against wavelength at the 60 nodes with soundings (median, with the 10th to 90th percentiles), and pure water's absorption a_w. K cannot physically fall below a_w.

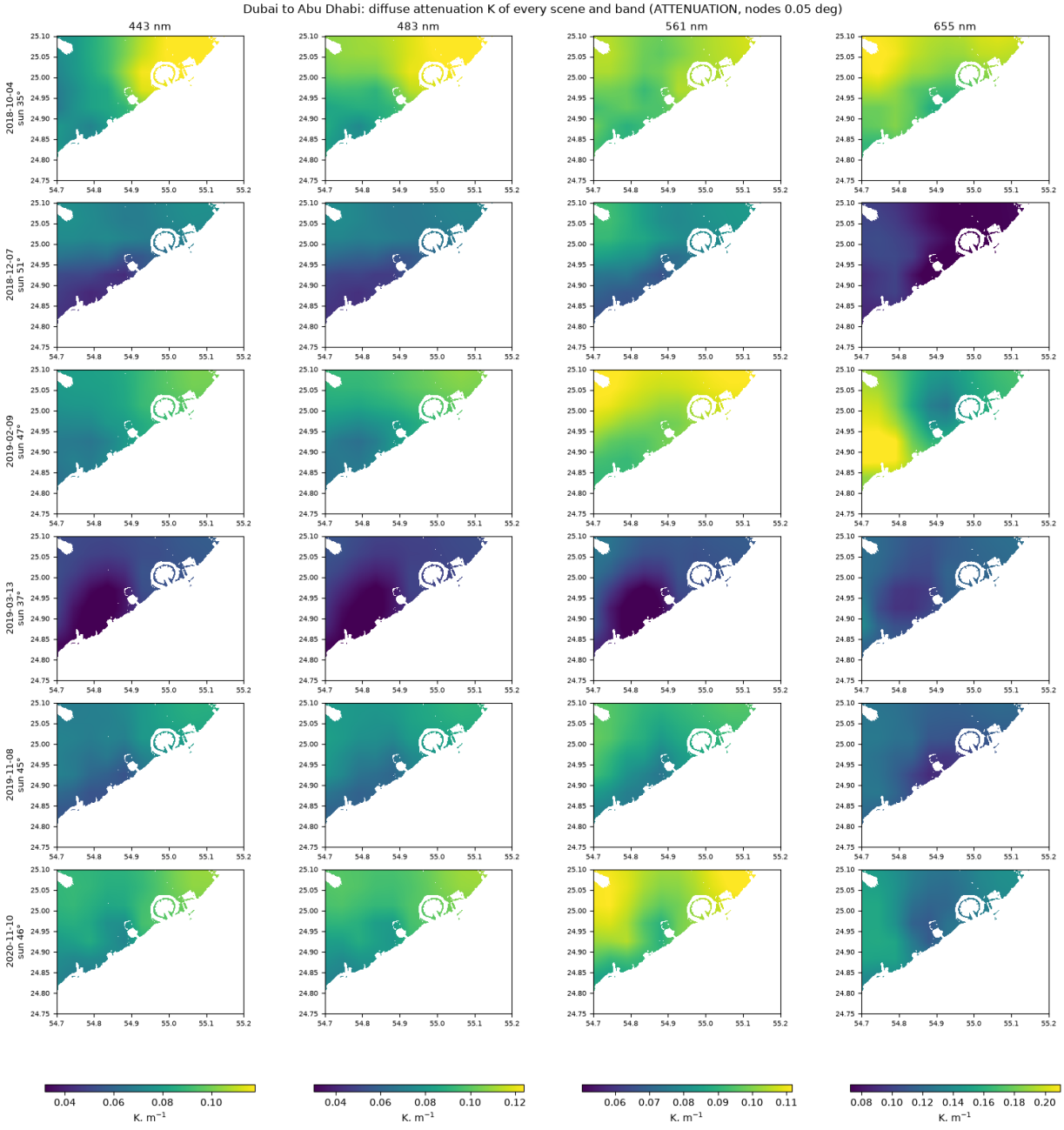


Figure 6: K of every scene (rows, with the date and sun zenith) and band (columns), on one colour scale per band. Nodes are 0.05° apart.

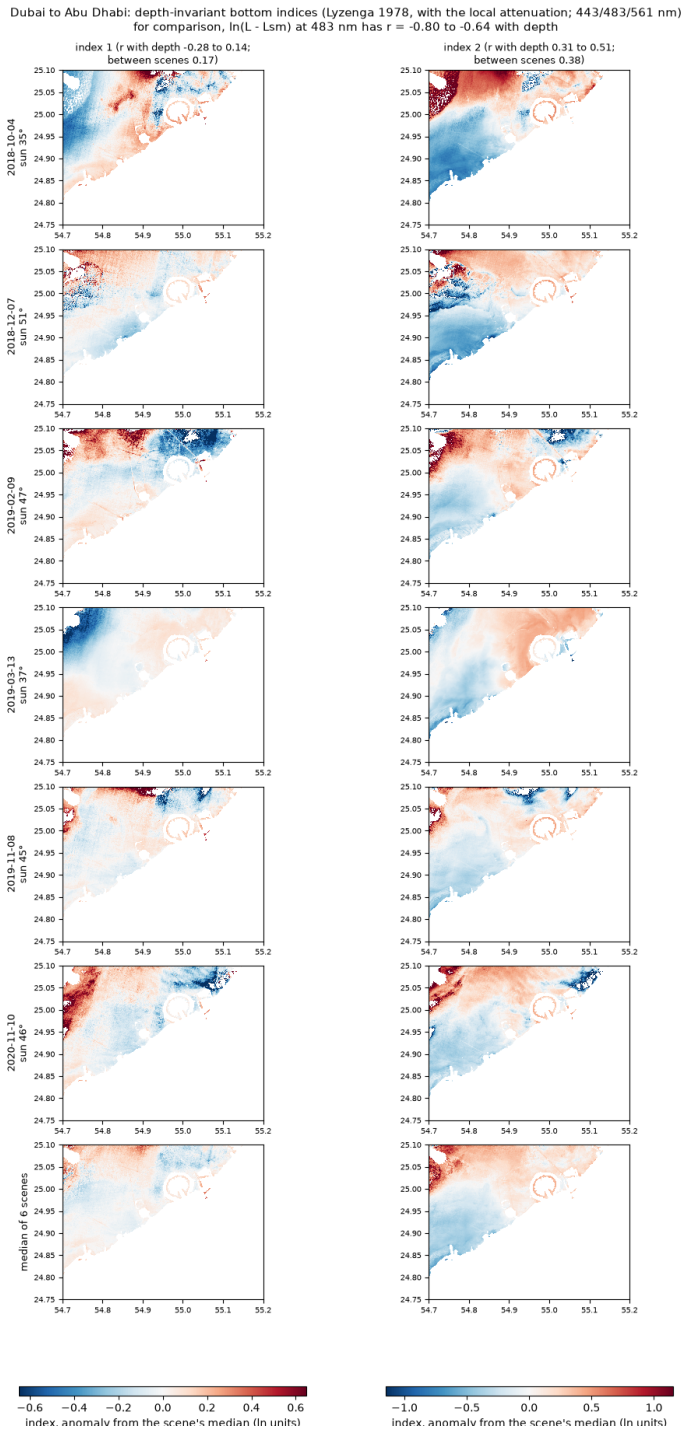


Figure 7: Depth-invariant bottom indices of every scene (443/483/561 nm, with the local attenuation), as anomalies from each scene's median, and their median over the scenes. The titles give each index's correlation with the ICESat-2 depth (six scenes) and between scenes (median).

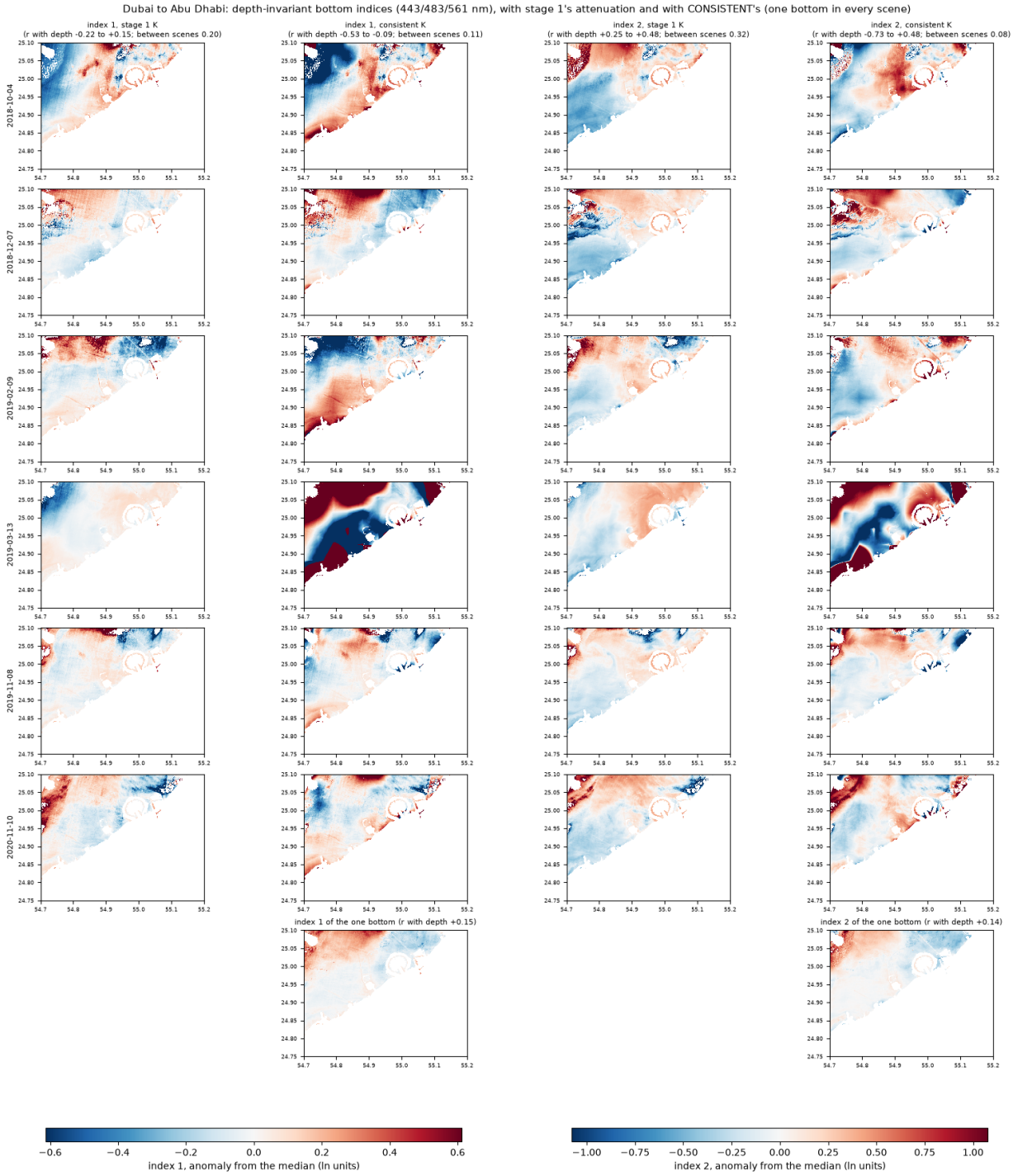


Figure 8: The depth-invariant indices (443/483/561 nm, fitted on the calibration half) of every scene, with stage 1's attenuation and with `CONSISTENT`'s, and the indices of `CONSISTENT`'s one bottom (bottom row), as anomalies from their medians. The titles give each index's correlation with the ICESat-2 depth (range over the scenes) and between scenes (median).